

GSST: Multimodal Graph-SMILES Fusion with Soft SMILES Tokens for Molecular Property Prediction

Jie Zhou, Qichang Zhao, Pengcheng Shu, Xiang Tang, and Jianxin Wang

Abstract—Accurate prediction of molecular properties is essential for drug discovery. While single modal molecular representations have shown promising results, their generalizability remains limited due to data sparsity and the inherent incompleteness of single modal characterizations. To overcome these limitations, we propose GSST, a novel multimodal pretraining framework that integrates graph-based molecular representations with SMILES sequences through learnable soft SMILES tokens. At the core of GSST is the G2S-Former module, which injects topological information from the graph-based representation into the soft SMILES tokens to enable effective cross-modal interaction while preserving modality-specific features. Extensive experiments on the MoleculeNet and MoleculeACE benchmarks demonstrate that GSST consistently outperforms state-of-the-art methods in molecular property prediction and activity cliff assessment. These results underscore the importance of effective multimodal alignment in capturing shared molecular patterns and alleviating the challenges posed by limited labeled data. GSST represents a scalable and high-throughput approach with significant potential to advance drug discovery.

Index Terms—Molecular Property Prediction, Multimodal Fusion, Activity Cliff Evaluation

I. INTRODUCTION

DRUG discovery is a complex and resource-intensive process, often requiring decades of time and billions of dollars in funding [1]. Accurate prediction of molecular properties is essential for rational drug design, with effective molecular representation serving as its foundation [2]. Recently, deep learning has emerged as a powerful tool for molecular representation learning and property prediction [3], and foundation models have further highlighted their transformative role in bioinformatics [4]. By capturing intricate relationships between molecular structures and their biological or

physicochemical properties, these methods facilitate efficient exploration of chemical space, thereby supporting data-driven drug discovery [5][6].

Current research utilizes two principal approaches for molecular representation: one-dimensional (1D) sequences and molecular graphs [7]. Common 1D representations include text-based notations and molecular fingerprints. The Simplified Molecular Input Line Entry System (SMILES) [8] is widely adopted in molecular learning tasks due to its efficiency. For 1D representations, recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) have been used. However, due to the success of the Transformer architecture in natural language processing, large-scale language models—initially designed for general language understanding and generation—have been adapted for molecular science applications [9]. ChemBERTa-2 [10] learns molecular representations by decoding randomly masked tokens in the input SMILES sequence. However, sequence-based approaches overlook 2D graph representations that are critical for understanding molecular structures.

To understand molecular graph structure, Graph Neural Networks (GNNs) treat molecules as graphs with atoms represented as nodes and chemical bonds as edges. These models aggregate topological information through various message-passing mechanisms [11]. GIN [12] combines neighbor features through learnable aggregation functions, and GAT [13] employs attention mechanisms to weight neighbor relationships. Additionally, DMPNN [14] and CMPNN [15] incorporate bond directionality for more accurate message passing. Concurrently, Zhang et al. [16] proposes a bond-centric re-wiring strategy that converts bonds into nodes and enforces consistency between the original and re-wired views. Similarly, Cai et al. [17] builds a dedicated bond-graph and performs contrastive learning between the atom-centric and bond-centric representations. Although recent single modal methods have alleviated some limitations of traditional unimodal approaches, they remain restricted to a single modality. For example, 1D methods capture sequential dependencies effectively but lack explicit structural awareness, whereas 2D methods improve structural encoding but ignore the sequential rules inherent in SMILES [18]. Consequently, they cannot fully exploit the complementary cues present in both sequential and graphical representations. This highlights the necessity of our synergistic fusion approach.

Molecular property prediction faces several major chal-

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For reproducibility, all code and data have been made available at <https://github.com/zoey731/GSST>.

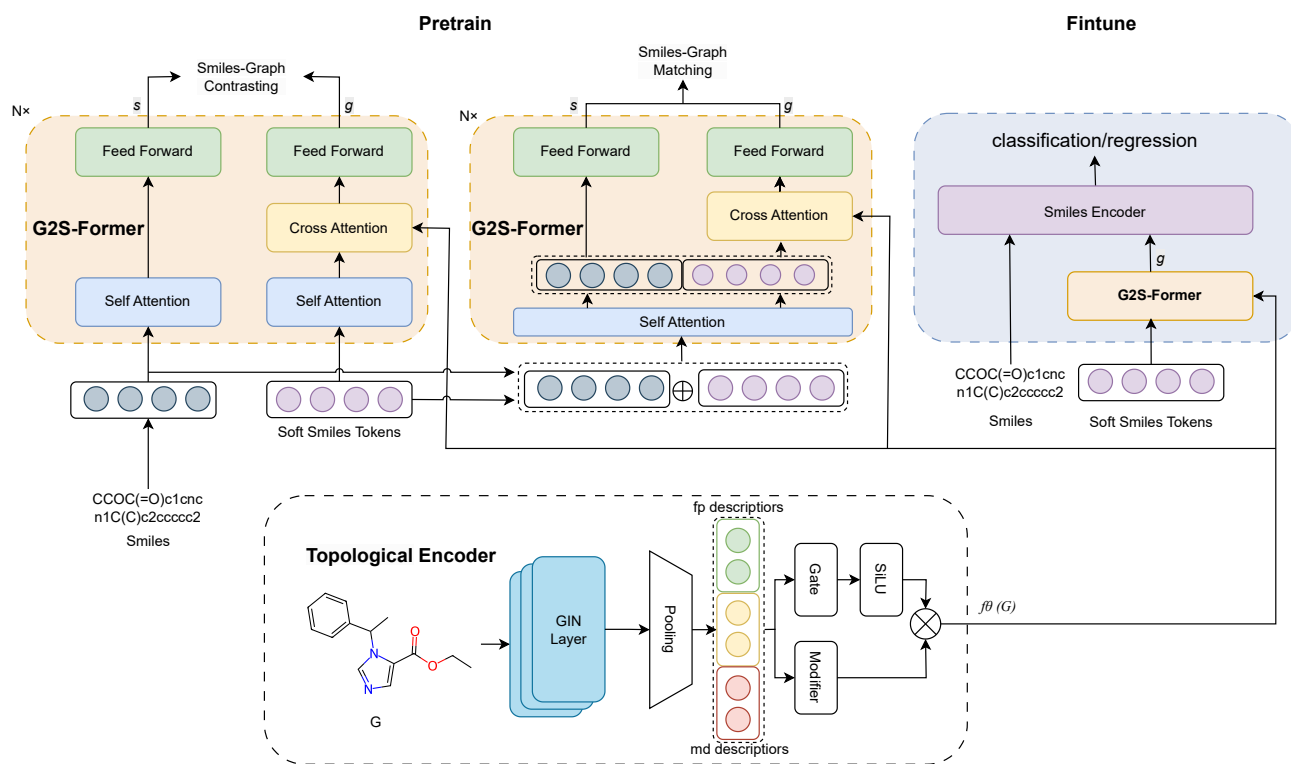


Fig. 1. Model Architecture. The GSST framework follows a pretrain-finetune paradigm. The topological encoder processes molecular structures while incorporating descriptors to build comprehensive representations. During pretraining, G2S-Former fuses topological features into soft SMILES tokens using contrastive and matching learning. For downstream tasks, the pretrained G2S-Former is fine-tuned for molecular property prediction.

Challenges: the scarcity of labeled data limits supervised learning effectiveness, and generalization remains difficult due to the vast chemical space and incomplete molecular representations [19][20]. To mitigate data scarcity, self-supervised learning methods leverage large scale unlabeled molecular data through pretraining to enable effective knowledge transfer. MolCLR [21] employs contrastive learning with graph augmentation strategies. Hu et al. [22] explores hierarchical pretraining at both node and graph levels. To obtain better molecular characterization, researchers have developed a multimodal model to simultaneously process SMILES sequences and graph structures. Wang et al. [23] combines multilevel message-passing neural networks for molecular graphs with SMILES sequences using weighted averaging. KPGT [24] employs a Line Graph Transformer with knowledge-guided pretraining to exploit structural and chemical knowledge. Similarly, MolGT [25] integrates multiview and multimodal data through a graph Transformer framework, demonstrating the potential of hybrid approaches. MolPROP [26] fuses ChemBERTa-2 language representations with graph neural networks for property prediction. MoleSG [27] leverages non-overlapping masked-language-and-graph pretraining to encourage fine-grained interaction between SMILES and molecular graphs. Beyond graph-sequence pairs, MM-FRL [28] presents a pretraining framework that unifies graphs, SMILES, images, fingerprints, and even NMR spectra via relational learning. Furthermore, GraphMVP [29] and MoleculeSDE [30] incorporate both 2D topological and 3D

geometric information. Likewise, Wang et al. [31] combines a 2D graph attention network with an equivariant 3D GNN to integrate spatial invariances.

Molecular pretraining techniques have demonstrated considerable efficacy in addressing the challenges posed by limited labeled data. Building upon this foundation, we hypothesize that effective alignment of multimodal molecular representations can facilitate the extraction of intrinsic molecular features. To enhance the interaction between 1D and 2D molecular representations, we propose GSST (Graph-SMILES Synergistic Transformer), a novel approach that leverages topological features to augment SMILES-based knowledge. First, molecular descriptor features are integrated into the graph representation via SwiGLU [32]. Through a G2S-Former module, the 2D topological features are subsequently transformed into soft SMILES tokens, enabling a unified representation of multimodal molecular features. Through pretraining and fine-tuning, GSST successfully integrates 1D sequence and 2D structural information while demonstrating particular sensitivity to molecular variations, as evidenced by its effective processing of activity cliff datasets. Comprehensive MoleculeNet benchmarks confirm GSST's superior performance across diverse physicochemical and biological property prediction tasks, highlighting its value for molecular representation learning and drug discovery applications.

II. METHODS

A. Overview

As shown in Figure 1, GSST adopts a pretraining and fine-tuning paradigm. Initially, the topological encoder module encodes the molecular structure, while descriptors are incorporated to build local and global representations of the molecular topology. During pretraining, G2S-Former fuses topological features into soft SMILES tokens, enabling unified multimodal representations. The model is optimized with two complementary objectives: a contrastive loss that enforces global alignment between modalities, and a matching loss that captures fine-grained token-level correspondence. Finally, the pretrained soft SMILES tokens and G2S-Former parameters are transferred to downstream tasks, where task-specific fine-tuning is conducted for molecular property prediction.

B. Model Architecture

1) *Topological Encoder*: First, we employ GIN [12] as our primary molecular graph encoder, which is a graph neural network built on a neighborhood aggregation framework. GIN performs well in molecular graph tasks because it can accurately capture structural information. GIN can both distinguish differences in molecular structures and preserve important functional details of molecules. We input atomic and bond features into a multilayer GIN network to get h_{graph} , the details of features are provided in Table I. However, traditional molecular graph encoders always focus on molecular structure and may overlook global physicochemical properties. To address this limitation, we augment the encoder by incorporating 200 molecular descriptors and 512-dimensional RDKit fingerprints [33] directly. These features include atom types, bond types, ring systems, topological structures, electronic properties, and pharmacophore features, among others. We concatenate the graph embeddings h_{graph} , molecular descriptors h_{dp} , and fingerprints h_{fp} into a unified representation:

$$h_{\text{concat}} = \text{Concat}(h_{\text{graph}}, h_{\text{dp}}, h_{\text{fp}}). \quad (1)$$

To optimize feature integration, we apply Swish-Gated Linear Units (SwiGLU) [32]:

$$\text{SwiGLU}(h_{\text{concat}}) = \text{SiLU}(h_{\text{concat}}W + b) \otimes (h_{\text{concat}}V + c), \quad (2)$$

where W, V, b, c are learnable parameters, $\otimes$ is the element-wise product. SwiGLU selectively extracts useful information from molecular topological structures and descriptors. It enhances critical molecular features while reducing noise and irrelevant data. This encoder boosts the model's expressive and generalization capabilities, enhancing its performance in molecular property prediction.

2) *G2S-Former*: In molecular representation learning, SMILES sequences are conventionally processed similarly to natural language: first tokenized into discrete symbols, then converted to continuous vector representations through token embeddings. However, this approach can effectively capture sequential patterns, conventional SMILES tokens inherently lack representation of molecular topology. To bridge graph and sequence domains, we introduce soft SMILES tokens—a hybrid representation that incorporates topological features via a learnable matrix $M = \{m_1, m_2, \dots, m_N\}$. These vectors

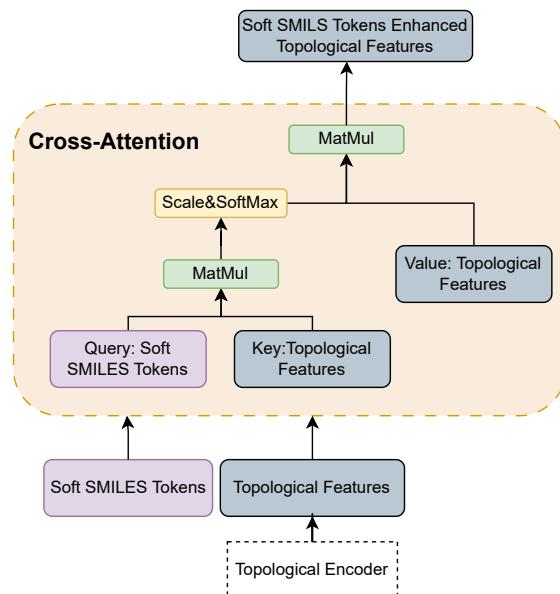


Fig. 2. Illustration of the cross-attention process in G2S-Former. Topological features are injected into soft SMILES tokens by using them as keys in the attention computation, resulting in the Soft SMILES Tokens-enhanced topological features.

are randomly Gaussian initialized and optimized jointly with the model. Within the G2S-Former, they interact with topological embeddings via cross-attention, evolving into a fixed-size set of Soft SMILES Tokens-enhanced topological features that are sequence-compatible and can be seamlessly integrated with SMILES tokens in the encoder.

From an information-theoretic perspective, the soft SMILES tokens can be regarded as selectively transferring complementary cues from the molecular graph into the SMILES space. Formally, while the SMILES sequence S and the molecular graph G both carry information about the prediction target Y , simple concatenation introduces redundancy due to their shared information $I(S; G)$. In contrast, the learnable soft SMILES tokens M are updated via cross-attention with graph embeddings in a way that encourages the maximization of the conditional mutual information $I(M; Y | S)$. This interpretation suggests that the tokens are optimized to retain graph features that are maximally predictive of the sequence.

Building on this design, we introduce the G2S-Former module, inspired by BLIP-2 [34], facilitates this integration through a dual-path architecture. Specifically:

SMILES branch: The SMILES branch of the G2S-Former, denoted as f_ϕ , is responsible for processing the sequence S . It first inserts a CLS token at the beginning and then processes it through multilayer self-attention modules and feedforward networks to generate a context-aware sequence embedding $s \in \mathbb{R}^d$.

$$s = \text{MeanPool}(f_\phi(S)). \quad (3)$$

Graph branch: For a molecular graph G , the topological encoder $f_\theta(G)$ extracts graph-based features. Cross-attention

TABLE I
ATOM AND BOND FEATURES IN THE ORIGINAL GRAPH.

Feature type	Features	Size
Atom		
Atom type	Types of atoms (e.g., C, N, O, F)	110
Degree	Number of bonds involving the atom	6
Atom mass	Atomic mass fraction	1
Is Aromatic	Is the atom in the aromatic group?	1
Hybridization	Sp,Sp2,Sp3,Sp3d,Sp3d2	5
TotalNumHs	Number of hydrogen atoms bonded to the atom	5
ChiralTag	Unspecified, Tetrahedral CW/CCW, or other	5
Formal Charge	Integer electronic charge assigned to an atom	5
Bond		
Bond type	Single, double, triple, aromatic bonds	4
Conjugated	Conjugation of bonds	1
Is Ring	Whether the bond is part of a ring	1
Stereo	None, any, E/Z or cis/trans	6

TABLE II
THE DESCRIPTION OF THE MOLECULENET DATASETS.

Task Type	Metric	Category	Dataset	Tasks	Compounds	Split Type
Classification	AUC-ROC	Biophysics	MUV	17	93087	scaffold
			HIV	1	41127	scaffold
			BACE	1	1513	scaffold
			BBBP	1	2039	scaffold
			Tox21	12	7831	scaffold
		Physiology	ToxCast	617	8575	scaffold
			SIDER	27	1427	scaffold
			ClinTox	2	1478	scaffold
			ESOL	1	1128	scaffold
			FreeSolv	1	642	scaffold
Regression	RMSE	Physical Chemistry	Lipophilicity	1	4200	scaffold
			FreeSolv	1	642	scaffold
			QM7	1	7160	scaffold
			QM8	12	21786	scaffold
	MAE	Quantum Mechanics	QM9	12	133885	random

TABLE III
THE DESCRIPTION OF THE MOLECULEACE DATASETS.

Type	Datasets	Description
GPCR	KOR, 5-HT1A, CB1, D4R, HRH1, MOR, D3R, DOR, HRH3, GHSR, OX2R	Important proteins involved in the process of cellular signal transduction.
Kinase	GSK3, ABL1, PIM1, JAK1, JAK2, CLK4	Regulate neural activity through potassium ion channels and protein kinase C signaling pathways.
NR	PPAR γ , PPAR α , AR, GR, FXR, PPAR δ	Participate in processes such as metabolism, development and reproduction.
Others	SERT, DAT, SOR	Influence on neural signal conduction.
Protease	F2, FX	Participate in blood coagulation.
Transferase	PIK3CA	It is closely related to the occurrence and development of various diseases.

layers fuse these features with soft SMILES tokens $M \in \mathbb{R}^{N \times d}$ producing a Soft SMILES Tokens-enhanced topological embedding g :

$$g = \text{CrossAttn}(f_{\theta}(G), \text{MultiHeadAttn}(M)). \quad (4)$$

To further clarify how topological features are injected into soft SMILES tokens, Figure 2 illustrates the cross-attention mechanism. Specifically, each soft SMILES token m_i first interacts with other tokens through self-attention, and is then used as a query vector in a cross-attention module with topological embeddings g_j , while the topological encoder outputs serve as keys and values. The updated token representations

are computed as:

$$\alpha_{ij} = \frac{\exp(\langle W_q m_i, W_k g_j \rangle)}{\sum_j \exp(\langle W_q m_i, W_k g_j \rangle)}, \quad (5)$$

where α_{ij} denotes the attention weight, W_q, W_k, W_v are learnable linear transformation matrices. Each soft SMILES token adaptively aggregates structural information from the molecular graph, thereby serving as a bridge that injects graph-level features into the SMILES encoder.

By inserting cross-attention every two layers, both local and global structural cues are progressively integrated into the soft SMILES tokens. Soft SMILES tokens and original SMILES

inputs share self-attention modules but employ separate feed forward networks to prevent over-mixing in deep layers. Ensures the model retains its sequential modeling strengths while benefiting from structural cues.

During fine-tuning in the downstream tasks, the raw SMILES tokens are first mapped to embeddings through SMILES encoder embedding layer. These embeddings are then concatenated with the soft SMILES tokens along the sequence dimension. The combined sequence of embeddings is then fed into the SMILES encoder, enabling the pretrained queries to inject topology-aware information into the sequence modeling process. Unlike traditional fusion methods such as simple concatenation or weighted averaging, G2S-Former enables structured, selective alignment between molecular graphs and SMILES sequences. This design yields richer, more flexible representations, leading to improved molecular property prediction.

C. Training Strategy

1) *Pretraining*: GSST proposes a pretraining framework to establish cross-modal alignment between molecular graphs and SMILES sequences via G2S-Former and soft SMILES tokens. During this stage, soft SMILES tokens serve as receptacles of topological information, progressively absorbing graph-derived features through the attention mechanism of G2S-Former. To ensure multimodal fusion, we design two complementary pretraining tasks: a contrastive learning task and a matching learning task.

The contrastive learning task improves the discriminative power of molecular representations by maximizing inter-modal similarity. This task strengthens global alignment across modalities, ensuring that the overall representations of the same molecule converge. For each molecule i , we derive a topology-aware representation g_i and a sequence representation s_i . For each matched pair (g_i, s_i) , we compute a bidirectional contrastive loss with a cosine similarity.

$$l_{g2s} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(g_i, s_i)/\tau)}{\sum_{j=1}^B \exp(\text{sim}(g_i, s_j)/\tau)}, \quad (6)$$

$$l_{s2g} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(s_i, g_i)/\tau)}{\sum_{j=1}^B \exp(\text{sim}(s_i, g_j)/\tau)}, \quad (7)$$

$$l_c = \frac{1}{2} l_{g2s} + \frac{1}{2} l_{s2g}, \quad (8)$$

where B is the batch size, τ the temperature, and $\text{sim}(s_i, g_i)$ in GSST is cosine similarity function. By construction, the numerator corresponds to the positive pair similarity, while the denominator contains all batch-wise negatives, making this a relative, ranking-based supervision signal. This encourages the model to learn a globally consistent embedding space where paired graph and SMILES representations converge.

In contrast, the matching objective adopts a pairwise binary classification formulation, predicting whether a given graph-SMILES pair matches. Here, the model directly estimates the probability $\rho(g, s) \in (0, 1)$ that a pair is aligned,

and is trained with a logistic loss:

$$l_m = -\frac{1}{B} \sum_{i=1}^B \left[\log \rho(g_i, s_i) + \mathbb{E}_{j \neq i} \log(1 - \rho(g_i, s_j)) \right], \quad (9)$$

where (g_i, s_i) is a positive pair and (g_i, s_j) are randomly sampled negatives. Unlike contrastive learning, this objective provides absolute supervision on each pair, without normalization across the batch. Thus, it emphasizes fine-grained matching correctness rather than global ranking.

Architecturally, matching learning employs a single self-attention layer between learnable soft SMILES tokens and raw SMILES sequences. This design allows each soft SMILES token to selectively attend to local SMILES symbols, enabling token-level cross-modal interactions. After pretraining, GSST can therefore capture and align topological motifs in the graph with their symbolic counterparts in the sequence.

2) *Fine-tune*: In downstream tasks, the pretrained G2S-Former and soft SMILES tokens are transferred to task-specific training. Specifically, the topological encoder processes the SMILES of target compounds to generate structural features. During fine-tuning, we only use the "Graph" branch of G2S-Former, which takes topological features and pretrained soft SMILES tokens as input, producing representations enriched with topological information. Then, the Soft SMILES Tokens-enhanced topological embeddings are concatenated with the original SMILES embeddings and treated at the input layer of SMILES encoder. We employ ChemBERTa-2 [10] as our SMILES encoder. Finally, the model ultimately performs predictions based on the combined information.

In addition, our SMILES encoder in fine-tuning can be replaced with other models that can handle SMILES sequences. The decoupled design of the model architecture gives the prediction module good versatility properties.

III. EXPERIMENTAL SETUP

A. Data Description

Our work employs PubChem324K [36] as the pretraining dataset. The dataset comprises 32,489 organic molecules with diverse structures which is widely used in cheminformatics, drug discovery, and biomedical research.

For downstream tasks, we use the MoleculeNet benchmark [37], which encompasses a variety of tasks across multiple domains including biophysics, physiology, physical chemistry, and quantum mechanics. As summarized in Table II, MoleculeNet includes 8 classification tasks and 6 regression tasks. We adopt a scaffold split (80/10/10 for train/valid/test) to ensure molecular scaffolds do not overlap across different sets. This provides a stricter evaluation of the model's ability to handle new molecular structures, aligning with real drug discovery needs.

We also evaluate our method on activity cliff datasets using the MoleculeACE benchmark [38]. It constructs the bioactivity data on 30 macromolecular targets collected from the ChEMBL dataset [39]. As detailed in Table III, the MoleculeACE benchmark includes various important target categories such as GPCRs, kinases, and nuclear receptors, covering multiple key biological processes such as cell

TABLE IV
MODEL PERFORMANCE ON CLASSIFICATION DATASETS

Classification(ROC-AUC%, higher is better $\uparrow$)									
model	BACE	BBBP	ClinTox	HIV	MUV	SIDER	Tox21	Toxcast	Rank
DMPNN ^a	85.3±5.3	71.2±4.9	90.5±2.1	75.0±2.1	76.2±2.8	63.2±2.3	68.9±1.3	63.7±0.2	6.3
GIN ^a	70.1±5.4	65.8±4.5	58.0±4.4	75.3±1.9	71.8±2.5	57.3±1.6	74.0±0.8	66.7±1.5	8.3
MoCLR _{GIN} ^a	82.8±0.9	73.3±1.0	89.8±2.7	77.4±0.6	78.9±2.3	61.2±3.6	74.1±5.3	65.9±2.1	5.0
PretriainGNN ^a	85.9±0.8	70.8±1.5	78.9±2.4	80.2±0.9	81.4±0.9	65.2±0.9	78.7±0.4	65.2±0.9	4.0
ChemBERTa-2	85.8±2.6	73.1±1.1	62.8±2.1	69.1±0.9	71.5±1.2	55.9±5.0	77.3±0.6	73.0±2.6	6.3
GraphMVP ^b	81.2±0.9	72.4±1.6	77.5±4.2	77.0±1.2	75.0±1.0	63.9±1.2	74.4±0.2	63.1±0.4	6.9
MoleculeSDE ^c	80.4±0.9	73.2±0.5	86.6±3.7	78.8±0.3	79.9±0.8	59.6±0.8	76.5±0.3	65.2±0.3	5.4
KPGT	87.8±0.5	89.6±1.5	86.9±1.7	62.5±0.9	78.4±2.8	63.4±3.5	75.1±0.0	71.1±0.2	7.3
MoIGT ^d	84.5±0.1	73.7±1.0	88.9±0.4	79.3±2.1	78.8±2.2	65.4±0.4	75.8±0.2	66.4±0.2	3.7
GSST	91.2±2.0	92.4±0.6	89.9±4.4	82.2±2.5	<u>79.9±3.0</u>	59.6±3.4	79.1±0.0	74.7±0.6	2.1

Bold indicates the best results, while underlined indicates the second-best results.

^a The results are from [35]. ^b The results are from [29]. ^c The results are from [30]. ^d The results are from [25].

TABLE V
MODEL PERFORMANCE ON REGRESSION DATASETS

Regression (RMSE/MAE, lower is better $\downarrow$)							
model	ESOL	FreeSolv	Lipophilicity	QM7	QM8	QM9	Rank
DMPNN ^a	0.98±0.26	2.18±0.45	0.65±0.05	105.8±13.2	0.0143±0.0002	0.00514±0.00001	3.2
GIN ^a	1.45±0.02	2.76±0.18	0.85±0.07	124.8±0.7	0.0371±0.0009	0.00824±0.00004	6.3
MoCLR _{GIN} ^a	1.11±0.02	2.30±0.15	0.79±0.01	87.2±2.0	0.0174±0.0013	0.00480±0.00003	3.8
PretriainGNN ^a	1.22±0.02	2.83±0.25	<u>0.74±0.00</u>	110.2±6.4	0.0191±0.0003	0.00922±0.00004	5.7
ChemBERTa-2	0.89±0.09	1.78±0.19	0.86±0.02	133.8±2.7	0.0273±0.0109	0.00517±0.00134	4.5
KPGT	0.98±0.11	<u>2.76±0.95</u>	0.60±0.03	<u>86.7±2.7</u>	0.0158±0.0021	0.00571±0.00001	3.5
GSST	<u>0.93±0.04</u>	1.38±0.18	0.85±0.00	84.9±3.3	0.0132±0.0001	0.00405±0.00002	1.8

Bold indicates the best results, while underlined indicates the second-best results.

^a The results are from [35].

TABLE VI
THE HYPER-PARAMETER SETTING OF GSST IN FINETUNING PROCESS

dataset	init lr	batch size	max epochs	weight decay
BACE	6.00E-05	32	100	3.00E-05
BBBP	4.00E-05	32	80	2.00E-05
ClinTox	9.00E-05	64	80	8.00E-05
SIDER	3.00E-05	32	100	7.00E-04
Tox21	6.00E-05	64	40	3.00E-05
Toxcast	6.00E-05	64	100	1.00E-07
HIV	1.00E-05	32	10	2.00E-05
MUV	2.00E-05	32	60	3.00E-06
ESOL	7.00E-05	32	80	5.00E-06
FreeSolv	9.00E-05	128	40	2.00E-05
Lipophilicity	2.00E-07	32	100	5.00E-05
QM7	4.00E-05	32	80	4.00E-05
QM8	4.00E-05	32	80	8.00E-05
QM9_random	7.00E-05	32	80	2.00E-06

signal transduction, metabolism, neural signal transduction, and blood coagulation. For each target on MoleculeACE, molecules are first clustered based on Extended Connectivity Fingerprints (ECFPs) using spectral clustering. Then, for each cluster, molecules are divided into training and test sets in an 8:2 ratio using stratified random sampling based on activity cliff labels.

B. Baselines

To ensure fair and effective experimental comparisons, we select baselines and categorize them into four groups: end-to-end models (i.e., DMPNN [14] and GIN [12]), graph-based pretrained models(i.e., PretriainGNN [22] and

MolCLR_{GIN} [35]), sequence-based pretrained models (i.e., ChemBERTa-2 [10]), and multimodal fusion methods. Multimodal fusion methods include GraphMVP [29] and MoleculeSDE [30], which integrate 2D and 3D information for pretraining, as well as KPGT [24] and MoIGT [25], both utilizing the transformer architecture to process graph-structured data. For KPGT and ChemBERTa-2, the original works do not report complete results on all downstream tasks considered in our study. To ensure fair comparison, we re-implemented their models using the same data split as GSST and optimized them with their own strategy.

For regression tasks, we use RMSE and MAE as evaluation metrics (MAE for QM7, QM8, QM9 datasets). For classifica-

tion, we use ROC-AUC. Our objective is to maximize ROC-AUC and minimize RMSE or MAE.

C. Implementation Details

The graph encoder is constructed by a 5-layer GIN. The number of soft SMILES tokens is set to 8 and initialized randomly. We set the maximum number of iterations to 100 for fine-tune and use an early stopping strategy (patience = 15) to prevent overfitting. We employ a Bayesian hyperparameter search based on the Tree-structured Parzen Estimator (TPE) [40]. By systematically exploring the parameter space of initial learning rate, batch size, maximum training epochs, and weight decay coefficient, the optimal hyperparameter combination is provided in Table VI.

IV. RESULTS AND DISCUSSION

A. Performance Comparison with Baselines

Table IV shows classification results, while Table V displays regression performance on MoleculeNet. To ensure a fair comparison, we follow the experimental setup of previous works [22], [29]. We repeat each experiment three times and report the mean and standard deviation of performance metrics on the test set. In this experiment, GSST demonstrates exceptional performance, achieving the best results in 5 classification tasks and 4 regression tasks. While GSST does not achieve SOTA on the ClinTox, MUV, and ESOL datasets, it still delivers competitive results.

GSST outperforms GIN and ChemBERTa-2 in classification tasks, demonstrating that integrating 1D SMILES sequences and 2D molecular graph structures achieves superior performance compared to unimodal molecular representation methods. Compared to KPGT and MolGT, which also employ transformer architectures for graph-structured molecular representation during pretraining, GSST achieves superior results, validating the effectiveness of our pretraining strategy. Specifically, fusing 2D features with soft SMILES tokens enables the accurate extraction of useful information from complex graph structures.

For the BBBP dataset, we construct a GSST variant by removing global descriptors while preserving graph features. GSST variant without global descriptors decreases performance by 9.1%, confirming the critical role of global feature integration. The results of all models on the SIDER dataset are not as expected. The potential reason is that drug side effects are influenced not only by molecular structure but also by various factors such as drug interactions. Relying only on the structure and textual representation of molecules is not sufficient to solve such complex biological problems.

On the QM7, QM8, and QM9 datasets, GSST demonstrates clear advantages over single-representation models using either graph-only or sequence-only molecular representations, emphasizing the need for richer molecular representations in quantum mechanics-related datasets.

B. Activity Cliffs Evaluation

We further evaluate our model on a more challenging activity cliff dataset. Activity cliffs refer to the phenomenon where

chemically similar molecules show significant differences in biological activity. Many minor structural changes, such as replacing a single atom or slightly modifying a functional group, can lead to a substantial increase or decrease in binding affinity to target proteins. This phenomenon is common in quantitative structure-activity relationship (QSAR) [41] studies, demanding models with high sensitivity to accurately capture these subtle structural variations.

We compare GSST with five baselines: four graph-based deep learning methods (GCN, MPNN, GAT, and AttentiveFP) and a transformer-based approach, which are selected from MoleculeACE. The transformer-based approach employs attention mechanisms and positional embeddings for sequence processing, pretrained on 10M compounds for bioactivity prediction. This selection is motivated by GSST's graph transformer architecture, allowing comprehensive evaluation against both conventional graph neural networks and contemporary transformer-based methods for molecular representation learning.

As shown in Fig. 3 and Table VII, we present the quantitative evaluation results across all molecules and specifically on activity cliffs within each test set. GSST outperforms baselines on 24 out of 30 datasets for general prediction (5.2% overall relative improvement) and demonstrates even better performance on activity cliffs, surpassing baselines on 25 out of 30 datasets (7.4% overall improvement). The results show that GSST outperforms the benchmark method in both the data sets of all molecules and those of molecules with only active energy barriers, achieving more significant improvements with smaller variances. It demonstrates robust predictive capability for molecular bio-activities, including accurate prediction of activity cliffs where it captures the impact of subtle structural changes on molecular activity. These observations collectively demonstrate GSST's efficacy in predicting molecular bio-activities, even when handling challenging cases like activity cliffs.

Fig. 4 shows the Euclidean distance of the active cliff pairs representations of different models. We select datasets for five distinct receptor types (GPCR, Transferase, Protease, Nuclear Receptor, and Kinase) from downstream tasks. We proceed to encode the active cliff molecule pairs with GSST, Transformer, GCN, GAT, MPNN, and AttentiveFP, ultimately computing the decision feature vectors (previous layer of output) of the Euclidean distance. This approach aims to assess whether molecules with similar structures can generate more different molecular representations, thereby completing the prediction of activity. The mean Euclidean distance computed by GSST is generally higher than other models, indicating a stronger discriminative capability. Taking the ChEMBL2147_Ki dataset as an example, although the mean value of GSST and the Transformer model are relatively close, the t-test results show that the difference between them is statistically significant ($p = 3.32 \times 10^{-2} < 0.05$), demonstrating that GSST still performs significantly better on this dataset.

C. Ablation Study

To comprehensively validate the effectiveness of GSST, we conduct ablation studies focusing on two critical aspects: the

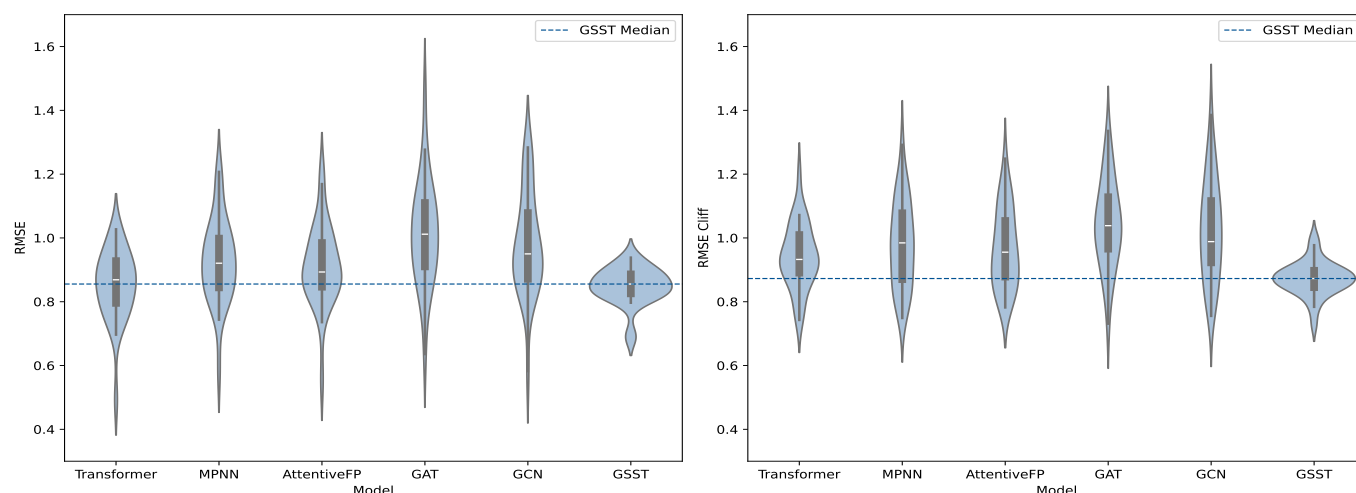


Fig. 3. Activity Cliffs Results on MoleculeACE. The averaged results of GSST and baseline methods on the 30 bioactivity datasets from MoleculeACE, measured in terms of RMSE on all molecules from the test sets (RMSE) and solely on activity cliffs from the test sets (RMSEcliff), respectively. The dotted line represents the median result of GSST (RMSE/RMSEcliff, which is used for intuitive comparison with other models).

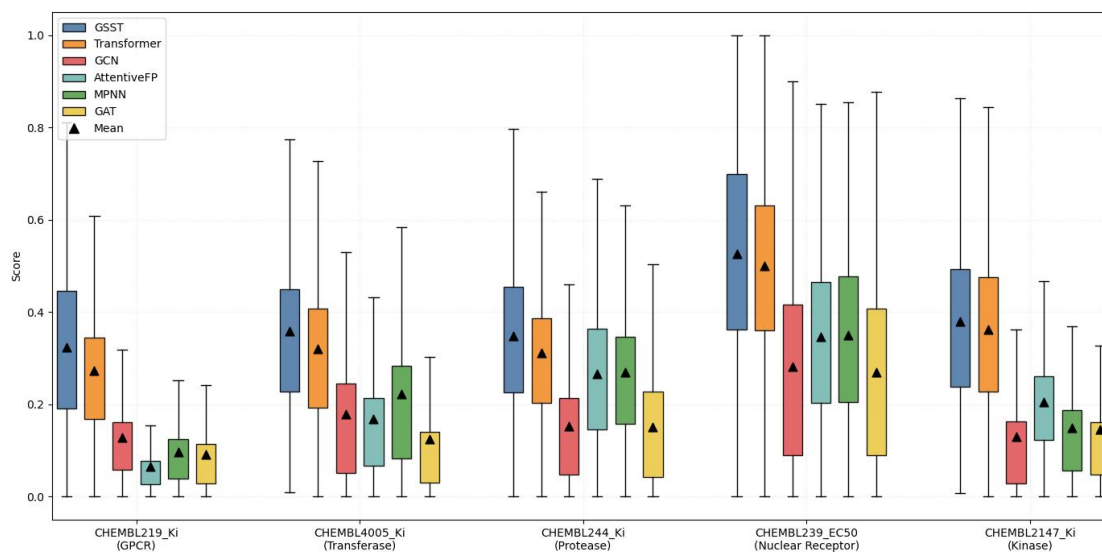


Fig. 4. The Euclidean distance of the active cliff molecule pairs across five receptor types (GPCR, Transferase, Protease, Nuclear Receptor, and Kinase). The triangular markers indicate the mean values for each model.

pretraining task framework and the feature fusion mechanism. Our pretraining framework combines contrastive learning and matching learning tasks. The contrastive learning component enhances molecular representation discriminability by maximizing feature similarity between positive sample pairs, while the matching learning component optimizes fine-grained semantic alignment. As shown in Fig. 5, by incrementally removing pretraining tasks (first retaining only contrastive learning, then only matching learning, and finally complete pretraining removal), we demonstrate that the complete pretraining strategy improves molecular property prediction performance, particularly for classification tasks.

For the feature fusion mechanism, we evaluate the soft SMILES tokens design which effectively bridges molecular graph features with SMILES sequences. We evaluate the impact of soft SMILES token quantity on model performance using four configurations (4, 8, 16, and 32 tokens). As shown

in Fig. 6, the model exhibits optimal performance across all tasks when configured with 8 tokens, indicating this as the most effective configuration. Furthermore, we directly input the molecular topological representation into the SMILES encoder. As can be seen from the dotted lines in Fig. 6, it conclusively demonstrates that soft SMILES tokens provide superior capability for learning and integrating molecular graph features. The above ablation experiments illustrate the validity and necessity of the GSST model design.

D. Visualization Analysis

To intuitively demonstrate the features learned by GSST, we select the BBBP dataset for classification task and the Lipophilicity dataset for regression task. For each dataset, we identify the seven most frequently occurring molecular scaffolds. To assess whether molecules with the same scaffold

TABLE VII
PERFORMANCE OF DIFFERENT MODELS ON ACTIVITY CLIFF DATASETS.

Datasets (ChEMBL ₁)	MoleculeACE						MoleculeACE_cliff					
	Transformer	MPNN	AttentiveFP	GAT	GCN	GSST	Transformer	MPNN	AttentiveFP	GAT	GCN	GSST
1862_Ki	0.961	0.948	1.347	0.987	0.942	0.825	0.775	0.749	0.808	0.731	0.755	0.876
1871_Ki	0.809	1.058	1.143	0.798	0.769	0.796	0.882	0.817	0.830	0.845	0.817	0.899
2034_Ki	0.804	0.905	0.931	0.809	0.810	0.691	0.913	1.009	0.970	0.999	0.989	0.905
204_Ki	1.098	1.458	1.553	1.138	1.056	0.900	1.072	1.167	1.165	1.268	1.386	0.914
2047_EC50	0.767	1.030	0.970	0.840	0.797	0.759	0.743	0.807	0.795	0.897	0.840	0.841
214_Ki	0.862	1.183	1.083	1.051	1.007	0.810	0.887	0.997	1.018	1.136	1.104	0.844
2147_EC50	0.903	1.025	1.906	0.966	0.840	0.927	0.933	0.915	0.956	1.048	1.062	0.861
216_EC50	0.869	1.053	1.046	0.957	0.928	0.856	0.930	0.988	0.958	0.970	0.982	0.897
219_Ki	0.878	0.903	0.952	0.979	1.026	0.966	0.937	0.981	1.020	1.139	1.122	0.904
228_Ki	0.908	1.000	1.192	1.026	0.958	0.869	0.990	1.000	0.938	1.035	1.139	0.871
231_Ki	0.964	1.305	1.262	0.991	0.878	0.816	1.048	0.832	0.882	1.197	0.839	0.815
233_Ki	1.072	1.074	1.211	1.066	1.056	0.882	0.967	1.077	0.998	1.042	1.098	0.907
234_Ki	0.863	0.959	0.855	0.950	0.934	0.855	0.927	0.865	0.857	0.969	0.945	0.844
235_EC50	0.801	1.058	1.202	0.869	0.901	0.795	0.899	1.049	1.058	1.092	1.077	0.839
236_Ki	1.024	1.364	1.370	1.002	0.942	0.888	1.053	1.158	1.127	1.215	1.248	1.001
237_EC50	1.126	1.402	1.361	1.103	1.132	0.939	1.024	1.124	1.136	1.095	1.164	0.978
237_Ki	0.996	1.053	1.310	1.085	1.112	0.889	0.962	1.115	1.143	1.249	1.249	0.939
238_Ki	0.880	1.142	1.216	0.928	0.937	0.824	0.898	0.888	0.954	1.022	0.998	0.831
239_EC50	0.910	1.288	1.361	0.902	0.906	0.873	0.932	1.080	1.083	1.108	1.083	0.865
244_Ki	1.078	1.660	1.706	1.088	1.075	0.896	1.071	1.177	1.044	1.336	1.246	0.898
262_Ki	0.976	1.021	1.116	0.994	0.934	0.862	1.018	0.855	0.915	0.963	0.914	0.826
264_Ki	0.822	1.082	1.102	0.896	0.855	0.814	0.819	0.873	0.782	0.915	0.826	0.777
2835_Ki	0.485	0.668	0.747	0.555	0.505	0.590	0.943	0.962	0.897	1.017	0.988	0.729
287_Ki	0.869	0.927	1.149	0.947	0.886	0.857	0.960	0.998	0.875	1.081	0.980	0.902
2971_Ki	0.826	0.973	1.091	0.803	0.781	0.853	1.070	1.123	1.104	1.047	1.088	0.883
3979_EC50	0.834	1.183	1.080	0.923	0.812	0.973	0.858	0.970	0.883	0.963	0.976	0.867
4005_Ki	0.855	0.998	1.059	0.861	0.875	0.845	0.864	0.799	0.831	0.853	0.884	0.784
4203_Ki	0.959	1.056	1.062	1.004	0.975	0.923	1.196	1.292	1.249	1.170	1.243	0.946
4616_EC50	0.784	0.935	0.947	0.873	0.867	0.780	0.773	0.810	0.833	0.860	0.832	0.853
4792_Ki	0.912	1.122	1.233	1.004	0.923	0.826	0.903	0.923	0.892	0.990	0.944	0.866
Average	0.896	1.004	1.186	0.947	0.914	0.849	0.942	0.980	0.967	1.042	1.027	0.872

Both columns report RMSE: MoleculeACE for the entire test set, and Molecule_cliff for the activity cliff subset only.

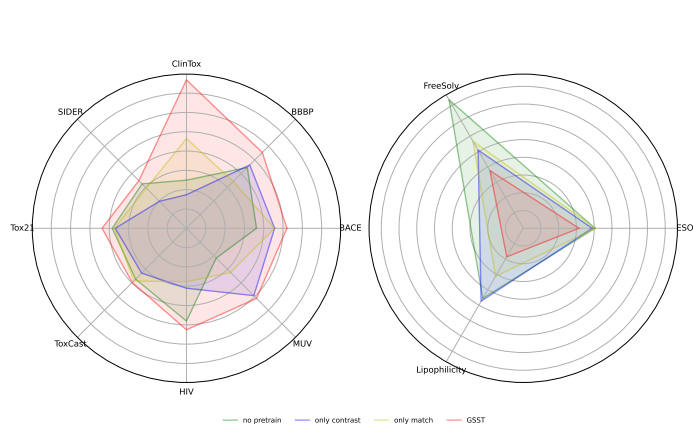


Fig. 5. Ablation experiment results on classification (left) and regression (right) tasks. The model performance is evaluated by ROC-AUC and RMSE.

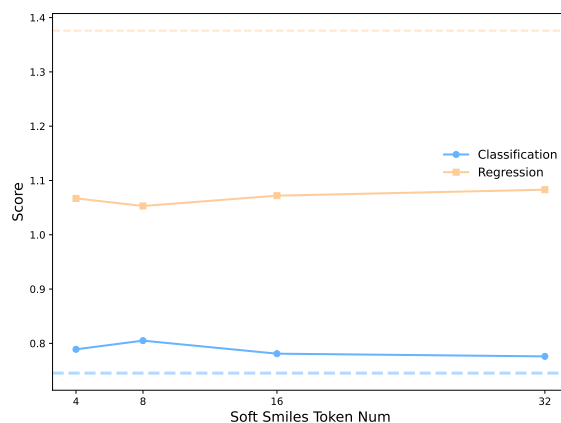


Fig. 6. The Effects of Soft Smiles Tokens. The blue line represents the classification result (ROC-AUC), while the yellow line represents the regression result (RMSE). The dotted line indicates that the graph features are directly concatenated to the SMILES encoder.

fold exhibit similar representation, we apply the t-SNE algorithm [42] to both pretrained and no-pretrain molecular representations, mapping these features into a 2D space. As shown in Fig. 7, no pretrain model struggles to effectively distinguish between molecules with different scaffolds. In contrast, GSST produces more significant clustering results. It proves GSST can better differentiate the features of molecules with different scaffolds. To further quantify the model's performance, the

study employs the Davies-Bouldin Index (DBI) [43] to evaluate the clustering results, where a lower index indicates better performance. The DBI also demonstrates the same conclusion. The results further validate that pretrained component of GSST can enhance molecular feature discrimination.

V. CONCLUSION

In this study, we propose GSST, which enables language models to learn topological features from molecular graphs

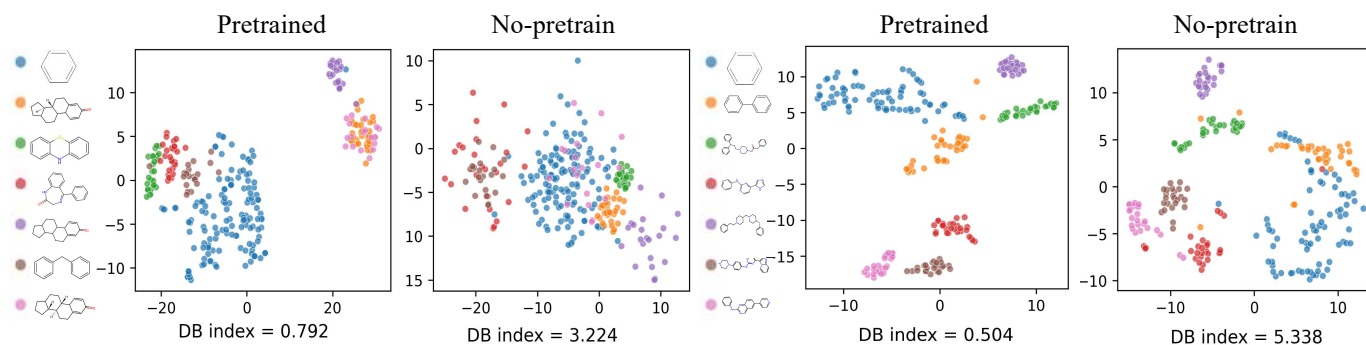


Fig. 7. Visual Molecular Representation Comparison. The left part shows the result of the BBBP dataset and the right shows the result of the Lipophilicity dataset. The corresponding DBI values are displayed below each subfigure.

while retaining their inherent strengths in SMILES. Specifically, we encode molecular topological structures using GIN for local features and descriptors for global features. We employ SwiGLU to filter and optimize the integration of molecular graph features and molecular descriptors. Then, we use G2S-Former to convert the molecular topological encoding into soft SMILES tokens, which can dynamically extract key topological structures to match SMILES rules and achieve fine-grained alignment between the topological domain and SMILES sequences. Compared to traditional multimodal fusion methods (i.e., concatenation or weighted averaging), GSST's cross-modal interaction mechanism demonstrates superior performance on molecular property prediction benchmarks (MoleculeNet) and activity cliff evaluation datasets (MoleculeACE). Ablation studies further validate the effectiveness of each component. GSST's robust performance on clinically relevant datasets, such as toxicity prediction (Tox21) and blood-brain barrier penetration (BBBP), underscores its potential for identifying novel drug candidates with desired properties.

Despite its advancements, GSST has certain limitations. First, the reliance on 2D molecular representations may hinder its ability to predict properties that depend on 3D spatial conformations. Addressing this issue is an important direction for future work. Specifically, GSST could be extended by incorporating 3D descriptors (i.e., molecular volume, surface area, molecular radius, principal moments of inertia, or radial distribution functions) derived from single or multiple conformers generated by RDKit. These descriptors could be added as global features, and the graph encoder may be replaced with 3D-aware graph neural networks. Furthermore, future studies may explore pretraining soft SMILES tokens in a 3D-enhanced setting, enabling a richer and more stereochemically informed representation for downstream tasks. Second, while soft SMILES tokens enhance multimodal fusion, their interpretability remains a challenge. Future research can therefore focus on two key objectives in molecular representation learning.

GSST's key innovation lies in its combination of graph-based and sequence-based representations through learnable soft SMILES tokens. This design not only improves predictive accuracy but also offers flexibility for downstream applica-

tions. From a clinical perspective, GSST's ability to detect small structural changes that significantly affect molecular activity (activity cliffs) is useful for drug discovery. This can help researchers improve potential drug candidates and reduce failures in later development stages. By combining different types of molecular data, this work supports more efficient drug discovery approaches.

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